**Code No: 157EP** **R18**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B. Tech IV Year I Semester Examinations, January/February - 2023**
**ENGINEERING MATERIALS**
**(Common to EEE, CSE)**

**Time: 3 Hours** **Max.Marks:75**

**Note:** i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

**PART – A**

**(25 Marks)**

| | | |
|---|---|---|
| 1.a) | What is the role of martensite in dual phase steel? | [2] |
| b) | What does TRIP stand for in steel? Give its applications. | [3] |
| c) | Name a few examples for super alloys applicable for high temperature applications. | [2] |
| d) | Distinguish between low and high temperature thermo mechanical treatment. | [3] |
| e) | What do you mean by extrinsic semiconductor? | [2] |
| f) | Mention the properties of piezoelectric materials. | [3] |
| g) | What is the significance of 18/8 stainless steel? | [2] |
| h) | Distinguish between austenitic and martensitic stainless steel. | [3] |
| i) | What are the types of biomaterials? | [2] |
| j) | Why titanium and its alloys are currently used as bio materials? | [3] |

**PART – B**

**(50 Marks)**

| | | |
|---|---|---|
| 2.a) | Give the composition, properties and applications of HSS steels? | |
| b) | Write short notes on dual phase steels. | [5+5] |
| | **OR** | |
| 3.a) | Explain the structure and properties of TRIP steels. | |
| b) | List the types and their typical applications of maraging steels? | [5+5] |
| 4.a) | Explain Topologically Close Packed (TCP) phase present in nickel based super alloys. | |
| b) | Discuss the properties and applications of Fe based super alloys. | [5+5] |
| | **OR** | |
| 5.a) | Discuss the main applications for single crystal super alloys. | |
| b) | Explain the effect of thermo-mechanical treatment on the microstructure and mechanical properties of a nickel based super alloys. | [5+5] |

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6.a) Differentiate between ferroelectric and piezoelectric materials.
b) Explain the properties of superconducting materials. [5+5]

**OR**

7.a) Explain about superconductivity materials which will be used at higher critical temperatures.
b) Give the typical applications of pyro-electric materials. [5+5]

8.a) Mention different grades of stainless steels. Briefly explain about martensitic stainless steel.
b) Explain the properties and applications of austenitic stainless steel. [5+5]

**OR**

9.a) Discuss the advantages and limitations of ferritic stainless steel.
b) Give the applications of stainless steels. [5+5]

10.a) What is biocompatibility and why is it so important?
b) Explain the engineering applications of biomaterials. [5+5]

**OR**

11.a) Discuss various important bio metallic alloys.
b) Mention various metallic materials used for biomedical applications. [5+5]

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